

Lecture 2: High-dimensional Spaces

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Motivation: Why study high-dimensional spaces?

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Modern data live in spaces of huge dimension – understanding their geometry is essential.

- Each data point (image, text, signal) is a vector in $\mathbb{R}^d$, often with $d \gg 1$.
- As d grows, **geometry and probability behave differently**: distances, volumes, and randomness no longer match our 3D intuition.

Roadmap of today's lecture

Main topics

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- **Geometry in high dimensions:** hypercubes and hyperspheres – what happens when d grows?
- **High-dimensional Gaussians:** a canonical random vector model.
- **Counterintuitive phenomena:** volume concentration, vanishing measure, and distances behaving strangely.

In high dimensions, geometry and probability intertwine – our intuition from 2D or 3D often fails.

Intuition from the High-Dimensional Cube

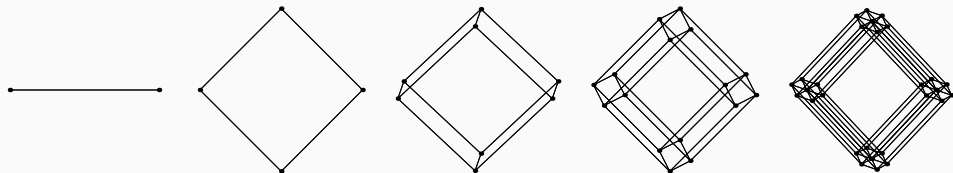


Figure 1: Hypercube in different dimensions. Source: Wolfram MathWorld.

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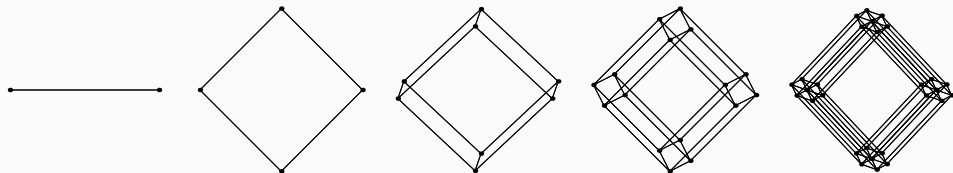


Figure 1: Hypercube in different dimensions. Source: Wolfram MathWorld.

First question

Draw two points x and y uniformly at random from the unit hypercube $[0, 1]^d$. How does their distance behave as the dimension d increases?

What feels “close” in 2D becomes “far” in high dimensions. Let’s quantify that.

Distance between two random points in a hypercube

- The squared Euclidean distance between two points is

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \sum_{i=1}^d (x_i - y_i)^2.$$

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- Note that $\|\mathbf{x} - \mathbf{y}\|_2^2$ can be viewed as the sum of d independent samples $z_i = (x_i - y_i)^2$ of a random variable z with **bounded variance**.

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As d increases, $\|\mathbf{x} - \mathbf{y}\|_2^2$ becomes a sum of many independent random contributions.

Concentration phenomenon in high dimensions

Law of Large Numbers

$$\mathbb{P}\left(\left|\frac{z_1 + z_2 + \cdots + z_d}{d} - \mathbb{E}[z]\right| \geq \varepsilon\right) \leq \frac{\text{Var}(z)}{d\varepsilon^2}.$$

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- As d increases, the right-hand side shrinks as $1/d$.
- Hence, the average distance per coordinate concentrates tightly around $\mathbb{E}[z]$.
- Consequently, the total squared distance $\|\mathbf{x} - \mathbf{y}\|_2^2$ concentrates around its mean, which scales on the order of $\mathcal{O}(d)$.

So random vectors in high dimension don't "spread out" uniformly – instead, they cluster on a thin shell around a sphere whose radius is determined by the mean squared distance

Intuition with the High-Dimensional Sphere

the hypersphere (or d -dimensional sphere) is a generalization of the circle (in dimension 2) or sphere (in dimension 3) to an arbitrary number of dimensions d :

Definition 1

The d -dimensional sphere is defined as a set of points $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$ such that $\sum_{i=1}^d x_i^2 = R$ where R is the radius of the hypersphere

Intuition with the High-Dimensional Sphere

we will see that the **volume of the hypersphere goes to zero** as the dimension increases and that the **maximum distance between two points stays constant**

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- recall that the geodesic distance between two points $\mathbf{x}$ and $\mathbf{y}$ on the sphere is given by
$$d(\mathbf{x}, \mathbf{y}) = r \arccos\left(\frac{\mathbf{x}^\top \mathbf{y}}{r^2}\right)$$
- for $r=1$, we have $d(\mathbf{x}, \mathbf{y}) = \arccos(\mathbf{x}^\top \mathbf{y}) \in [0, \pi]$.
- therefore, the distance does not depend on the dimension d .

Surface Area of High-Dimensional Sphere (notation)

Definition 2

we will denote by $\mathcal{S}^{d-1}(r)$ the d -dimensional sphere of radius $r > 0$ embedded in $\mathbb{R}^d$:

$$\mathcal{S}^{d-1}(r) = \left\{ \mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 = r \right\}$$

next slide we relate the surface area of a hypersphere of radius $r > 0$ to a unit hypersphere ($r = 1$)

Surface Area of High-Dimensional Sphere (rescaling)

Proposition 1 (rescaling to unit hypersphere)

the surface area of the d -dimensional sphere of radius r , denoted by $A_r(d)$ is equal to

$$A_r(d) = r^{d-1} A_1(d)$$

Surface Area of High-Dimensional Sphere (rescaling)

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hint: this is because

$$\int_{\mathcal{S}^{d-1}(r)} d\Omega(\mathbf{x}) = \int_{\mathcal{S}^{d-1}(1)} r^{d-1} d\Omega(\mathbf{x})$$

as the surface elements have dimension $d - 1$

Surface Area of High-Dimensional Sphere (formula)

we compute the surface area of the d -dimensional sphere which we denote by $A_r(d)$

Proposition 2 (surface area)

the surface area of the hypersphere $\mathcal{S}^{d-1}(r)$ is equal to

$$A_r(d) = \frac{2r^{d-1}\pi^{d/2}}{\Gamma(\frac{d}{2})}$$

- Gamma function: $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t} dt$
- as $z \rightarrow \infty$, $\Gamma(z) \simeq \sqrt{z}(z/e)^z$
- for fixed radius, $A_r(d) \rightarrow 0$ as $d \rightarrow \infty$

Surface area of High-Dimensional Sphere (proof)

proof:

let $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $f(\mathbf{x}) = e^{-\|\mathbf{x}\|_2^2}$. We will compute

$$\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} = \int_{\mathbb{R}^d} e^{-\|\mathbf{x}\|_2^2} d\mathbf{x}$$

in both Cartesian and polar coordinates:

$$\int_0^\infty e^{-r^2} \int_{\mathcal{S}^{d-1}(r)} d\Omega(\mathbf{x}) dr$$

Surface area of High-Dimensional Sphere (proof)

we have (Cartesian coordinates)

$$\begin{aligned}\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} &= \int_{\mathbb{R}^d} e^{-\|\mathbf{x}\|_2^2} d\mathbf{x} \\ &= \int_{\mathbb{R}^d} e^{-x_1^2} \dots e^{-x_d^2} dx_1 \dots dx_d \\ &= \left(\int_{-\infty}^{\infty} e^{-x_1^2} dx_1 \right) \dots \left(\int_{-\infty}^{\infty} e^{-x_d^2} dx_d \right) \\ &= \pi^{d/2}\end{aligned}$$

Surface area of High-Dimensional Sphere (proof)

next, we will perform the computation of the integral $\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x}$ in polar coordinates:

$$\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} = \int_0^\infty \int_{\mathcal{S}^{d-1}(r)} f(\mathbf{x}) d\Omega(\mathbf{x}) dr \quad (\text{coarea formula}),$$

where $d\Omega(\mathbf{x})$ is the surface area of the infinitesimal piece of the solid angle of the unit sphere

Surface area of High-Dimensional Sphere (finish)

noting that $f(\mathbf{x}) = e^{-\|\mathbf{x}\|_2^2} = e^{-r^2}$, and continuing with our calculation, we get

$$\begin{aligned}\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} &= \int_0^\infty \int_{\mathcal{S}^{d-1}(r)} e^{-r^2} d\Omega(\mathbf{x}) dr \\ &= \int_0^\infty e^{-r^2} \int_{\mathcal{S}^{d-1}(r)} d\Omega(\mathbf{x}) dr\end{aligned}$$

Note that $\int_{\mathcal{S}^{d-1}(r)} d\Omega(\mathbf{x})$ is the surface area $A_r(d)$ of the d -dimensional sphere. Therefore

$$\begin{aligned}\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} &= \int_0^\infty e^{-r^2} A_r(d) dr \\ &= A_1(d) \int_0^\infty e^{-r^2} r^{d-1} dr,\end{aligned}$$

where we used Proposition 1 in the last step

Surface Area of High-Dimensional Sphere

evaluating the remaining integral (we perform the change of variable $t = r^2$, i.e. $dt = 2rdr$) gives

$$\begin{aligned}\int_0^\infty e^{-r^2} r^{d-1} dr &= \int_0^\infty e^{-t} t^{(d-1)/2} \frac{1}{2\sqrt{t}} dt = \frac{1}{2} \int_0^\infty e^{-t} t^{d/2-1} dt \\ &= \frac{1}{2} \Gamma\left(\frac{d}{2}\right),\end{aligned}$$

where $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ is the Gamma function

we conclude that

$$\pi^{d/2} = \int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} = A_1(d) \frac{1}{2} \Gamma\left(\frac{d}{2}\right) = A_r(d) r^{-(d-1)} \frac{1}{2} \Gamma\left(\frac{d}{2}\right)$$

□

Volume of High-Dimensional Sphere (rescaling)

we start with a proposition that gives a relation between the volume of a hypersphere of radius r and the volume of the unit hypersphere:

Proposition 3 (rescaling to unit hypersphere)

the volume of the d -dimensional sphere of radius r , denoted by $V_r(d)$ is equal to

$$V_r(d) = r^d V_1(d)$$

Volume of High-Dimensional Sphere (formula)

Proposition 4 (volume)

the volume of the d -dimensional unit sphere is equal to

$$V_1(d) = \frac{1}{d} A_1(d) = \frac{\pi^{d/2}}{\frac{d}{2} \Gamma\left(\frac{d}{2}\right)}.$$

Remark 1

since $\Gamma\left(\frac{d}{2}\right)$ grows much faster than $\pi^{d/2}$ as function of d , we see that $V(d) \rightarrow 0$ as $d \rightarrow \infty$

Volume of High-Dimensional Sphere (proof sketch)

Proposition 5 (volume)

the volume of the d -dimensional unit sphere is equal to

$$V_1(d) = \frac{1}{d} A_1(d) = \frac{\pi^{d/2}}{\frac{d}{2} \Gamma(\frac{d}{2})}.$$

proof:

note that

$$V_1(d) = \int_{r=0}^1 \int_{\mathcal{S}^{d-1}(r)} d\Omega dr = \int_{r=0}^1 \int_{\mathcal{S}^{d-1}(1)} r^{d-1} d\Omega dr = \frac{1}{d} \int_{\mathcal{S}^{d-1}(1)} d\Omega = \frac{A_1(d)}{d},$$

where $A_1(d)$ is as given previously. □

The Volume of the High-Dimensional Sphere is Near the Equator (setup)

- consider the unit hypersphere with the north pole on the x_1 axis at $x_1 = 1$
- divide the sphere in half by intersecting it with the plane $x_1 = 0$
- the intersection of this plane with the sphere forms an object dimension $d - 1$: $\{\mathbf{x} \mid \|\mathbf{x}\| \leq 1, x_1 = 0\}$. We call it the **equator**

The Volume of the High-Dimensional Sphere is Near the Equator (main claim)

we will see that in high dimensions, most of the volume of the upper hemisphere is concentrated very close to the equatorial plane $x_1 = 0$

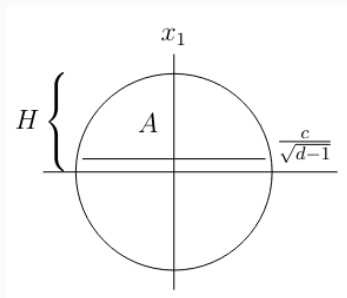


Figure 2: Most of the (shrinking) volume of the upper hemisphere of the d -dimensional ball is close to the equator

Define the upper cap T

to see this, let's define

$$T = \{\mathbf{x} \mid \|\mathbf{x}\| \leq 1, x_1 \geq \epsilon\}$$

as the portion of the d -dimensional hypersphere above the plane $x_1 = \epsilon$:

- each slice at height x_1 is a $(d-1)$ -dimensional hypersphere of radius $\sqrt{1-x_1^2}$
- The volume of this hypersphere is $(1-x_1^2)^{(d-1)/2} V_1(d-1)$ (Proposition 3)
- Hence

$$V^{(T)} = V_1(d-1) \int_{\epsilon}^1 (1-x_1^2)^{(d-1)/2} dx_1.$$

Bounding the Volume Above $x_1 = \varepsilon$

The integral is difficult to compute but one can derive the following upper bound on $V^{(T)}$:

$$V^{(T)} \leq \frac{1}{\epsilon(d-1)} e^{-\frac{d-1}{2}\epsilon^2} V_1(d-1)$$

(see A. Blum, J. Hopcroft & R. Kannan, Foundations of Data Science, p.20)

Lower bound for the hemisphere

next, we lower bound the volume of the entire upper hemisphere $V^{(U)}$ by the volume below the plane $x_1 = \frac{1}{\sqrt{d-1}}$

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next, we lower bound the volume of the entire upper hemisphere $V^{(U)}$ by the volume below the plane $x_1 = \frac{1}{\sqrt{d-1}}$

- to do so, we integrate the slabs between $x_1 = 0$ and $x_1 = \frac{1}{\sqrt{d-1}}$, which is at least the volume of the cylinder of radius $\sqrt{1 - \frac{1}{d-1}}$ and height $\frac{1}{\sqrt{d-1}}$

Lower bound for the hemisphere

- the volume of the cylinder is $\frac{1}{\sqrt{d-1}}$ times the $(d-1)$ -dimensional volume of the hypersphere $R = \left\{ \mathbf{x} \mid \|\mathbf{x}\| \leq 1, x_1 = \frac{1}{\sqrt{d-1}} \right\}$ (denote the volume of the hypersphere by $V^{(R)}$)
- since R is a $(d-1)$ -dimensional hypersphere of radius $\sqrt{1 - \frac{1}{d-1}}$, its volume is equal to

$$V^{(R)} = V_1(d-1) \left(1 - \frac{1}{d-1}\right)^{(d-1)/2} \geq V_1(d-1) \left(1 - \frac{1}{d-1} \frac{d-1}{2}\right) = \frac{1}{2} V_1(d-1),$$

where we used $(1-x)^a \geq 1-ax$

Ratio of volumes

- Therefore, the volume of the entire upper hemisphere satisfies

$$V^{(U)} \geq \frac{1}{2\sqrt{d-1}} V_1(d-1)$$

- Since

$$V^{(T)} \leq \frac{1}{\epsilon(d-1)} e^{-\frac{d-1}{2}\epsilon^2} V_1(d-1),$$

combining the bounds on $V^{(T)}$ and $V^{(U)}$ yields

$$\frac{V^{(T)}}{V^{(U)}} \leq \frac{2}{\epsilon\sqrt{d-1}} e^{-\frac{d-1}{2}\epsilon^2}$$

Interpretation: Concentration near the equator

- For any arbitrarily small $\epsilon > 0$,

$$\frac{V^{(T)}}{V^{(U)}} \leq \frac{2}{\epsilon\sqrt{d-1}} e^{-\frac{d-1}{2}\epsilon^2} \rightarrow 0$$

as $d \rightarrow \infty$.

- Let $V^{(E)}$ denote the volume in the equatorial band (i.e., below $x_1 = \epsilon$ in the upper hemisphere). Then

$$\frac{V^{(E)}}{V^{(U)}} := \frac{V^{(U)} - V^{(T)}}{V^{(U)}} \rightarrow 1 \quad \text{as } d \rightarrow \infty,$$

meaning most of the (shrinking) volume is squeezed into a **thin equatorial band** around $x_1 = 0$.